Basic Economic Concepts

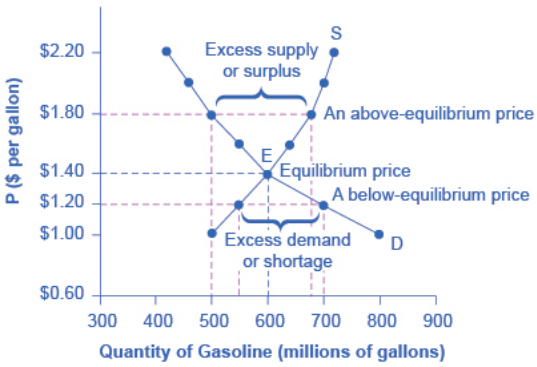
Introduction

- **Terminologies:**
  - **Opportunity Cost** - The cost of any action in terms of what you could have done instead
  - **Constrained Optimization**
    * Constrained -> Every decision has an `opportunity cost`
    * Optimization -> Make yourself as well off as possible
  - **Model**
    * To simplify things down *(strip down the variables e.g. Occam razor)*
    * Intuitively
    * Graphically
    * Mathematically

Supply and Demand

> Most economics model don't represent the reality instead the are trying approximate it

- **Supply and Demand Model**
  - **Demand -** How much people want it! *(The greater the demand, the higher the price)*
  - **Supply -** How much is available *(The greater the supply, the lower the price)*
  - **Equilibrium point -** The point in which both the consumer and the provider are happy to purchase and provides the goods *(This is where the "demand" curve intersects with the "supply" curve)*
    #+name: Equilibrium Point

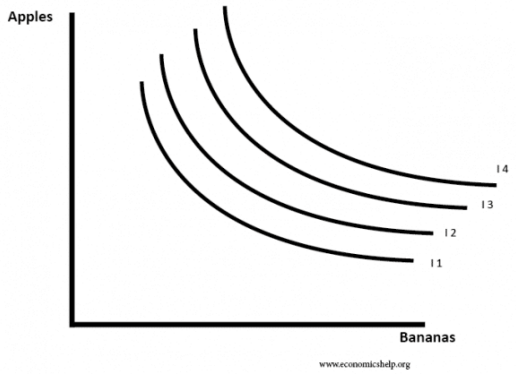


Market Economy vs Command Economy

- **Command Economy:**
  - Trying to even things out among people *e.g. Soviet Union economy*
  - Government decides the economy *(Too much to handle -> shortages, corruption, waiting list, etc)*
- **Market Economy:**
  - Don't try to even things out instead let,
  - Individual decides their own needs and production

Preferences

- **Preferences:**
  - Can be either `indifferent` or `prefer`
  - Logically consistent
  - More is better *(to some extend)*
  - `preference curve` - A condensed or simplify preference of some aspects
- **Properties of a map of indifference curve:**
  1. `consumer` prefer higher indifference curves
  2. Are downward sloping
  3. Never cross
  4. There is **one** indifference curve through each possible `bundle`
     #+name: Indifference Curves



- A level of `total ultility` represents the level of satisfaction when choosing a `bundle` -> *higher level of total utility leads to better satisfaction*

Utility Functions and Marginal Utility

- **Utility function -** A mathematical expression that translates bundles of goods into a single valuation of utility
- **Marginal Utility -** The change in utility from each additional good you consume

> With each extra unit of a good increase utility by less than the previous unit of the good this is called Diminishing Margin Utility (DMU)

> Different goods come with different level of DMU e.g. socks vs newspaper

Application - Soda pricing

> One of the best examples of diminishing marginal utility is how stores price different size of goods. It shows how `utility` and `supply-and-demand` works together

- Demand is high from **some** drink, but NOT so high for **additional** drink
  - On a hot day demand for a drinks is higher but demand for **additional** drink will be lower *-> Diminishing marginal utility in action*
  - Lower price could be charge for **additional** drink

The Math Behind Utility Functions

- There are lot of different `utility functions`
- A `utility function` should:
  - Be realistic
    * More is better
    * Have a `diminishing` component
  - Easy to work with
  - e.g. $U = \sqrt{(P * C)}$

Introduction to Budget Constraints